Rotation-Robust Bangla Sign Language Recognition from a Wrist-Referenced 6-IMU Glove

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***Abstract*—We present a low-cost wearable glove instrumented with six MPU6050 inertial measurement units (IMUs) — one per finger plus a sixth reference unit on the back of the hand — for isolated Bangla Sign Language (BdSL) recognition. The central design claim is that reporting each finger's orientation *relative to the wrist* ($R_{\text{rel}} = R_{\text{wrist}}^{\top} R_{\text{finger}}$) rather than relative to the Earth frame cancels whole-arm motion and yields a representation that is exactly invariant to global hand rotation. We verify this claim in three stages: (1) algebraically and numerically, on the logged data itself; (2) empirically, under a synthetic $SO(3)$ rotation stress test, across three model families (Random Forest, SVM, and a CNN–BiLSTM–attention network); and (3) as a label-efficiency question, via self-supervised masked-reconstruction pretraining. The rotation-robust representation loses 6–8 points of clean macro-F1 relative to absolute-orientation baselines, but exhibits *zero* degradation under rotation across all three model families, while absolute-orientation representations lose 0.82–0.87 macro-F1 under the identical stress test. A self-supervised pretraining pass on 24 unlabeled sessions is also attempted and is reported as a negative result: the pretext task is shown, via a no-learning baseline, to be too easy to justify any downstream claim built on it. All results are within-signer, on a single continuous recording session, and are reported with that scope stated explicitly.**

***Index Terms*—inertial measurement units, wearable sensing, sign language recognition, rotation invariance, sensor fusion, gesture recognition**

I. Introduction

Wearable inertial gloves are an established approach to sign-language and gesture recognition, avoiding the lighting, occlusion, and camera-framing failure modes of vision-based systems. A finger-mounted IMU alone, however, reports orientation against the Earth's gravity and magnetic frame, which conflates two physically distinct events: the finger flexing relative to the palm, and the entire arm moving in space. A classifier trained on such features risks learning arm pose as a shortcut rather than hand shape, a failure that would not surface until the signer's orientation relative to the sensor's calibration changes.

This project instruments a glove with a sixth, wrist-mounted IMU to serve as a body-frame reference, and computes each finger's orientation relative to that frame via full rotation-matrix composition rather than naive Euler-angle subtraction. The central empirical question is whether this representation choice actually delivers the invariance it is designed for, and whether that invariance survives contact with real classifiers rather than remaining a purely algebraic property.

TABLE I
TCA9548A CHANNEL ASSIGNMENT.

Sensor	Channel
Thumb	4
Index	3
Middle	6
Ring	7
Pinky	2
Wrist (reference)	5

A. Contributions

- A verified 6-IMU glove instrument (five fingers plus a wrist reference) with documented calibration, multiplexer-fault detection, and channel-mapping verification procedures.
- An algebraic and numerical proof that the firmware's relative-orientation columns implement true rotation composition, not angle subtraction.
- A three-model-family empirical test (Random Forest, SVM, CNN–BiLSTM–attention) showing that the wrist-relative representation achieves *exactly zero* macro-F1 degradation under a synthetic global rotation, while absolute-orientation representations lose 0.82–0.87.
- A quantified confound analysis of the underlying recording session, bounding how much of any accuracy figure could be explained by recording order alone.
- A negative result for self-supervised masked-reconstruction pretraining at this data scale, diagnosed rather than merely reported.

II. Instrument

A. Hardware

The glove carries six MPU6050 6-axis accelerometer/gyroscope modules multiplexed on a single I2C bus via a TCA9548A 8-channel multiplexer, read by an ESP32 microcontroller. Table I gives the channel assignment. Five sensors give hand shape; the sixth, mounted on the back of the hand, gives the body-frame reference.

The firmware streams CSV over serial at 921,600 baud. The measured effective sampling rate is 62.5–67.8 Hz (mean 63.4 Hz across sessions), short of the firmware's 100 Hz target; this was traced to I^2C read latency across six multiplexed sensors rather than to serial bandwidth, which has ample headroom at this baud rate.

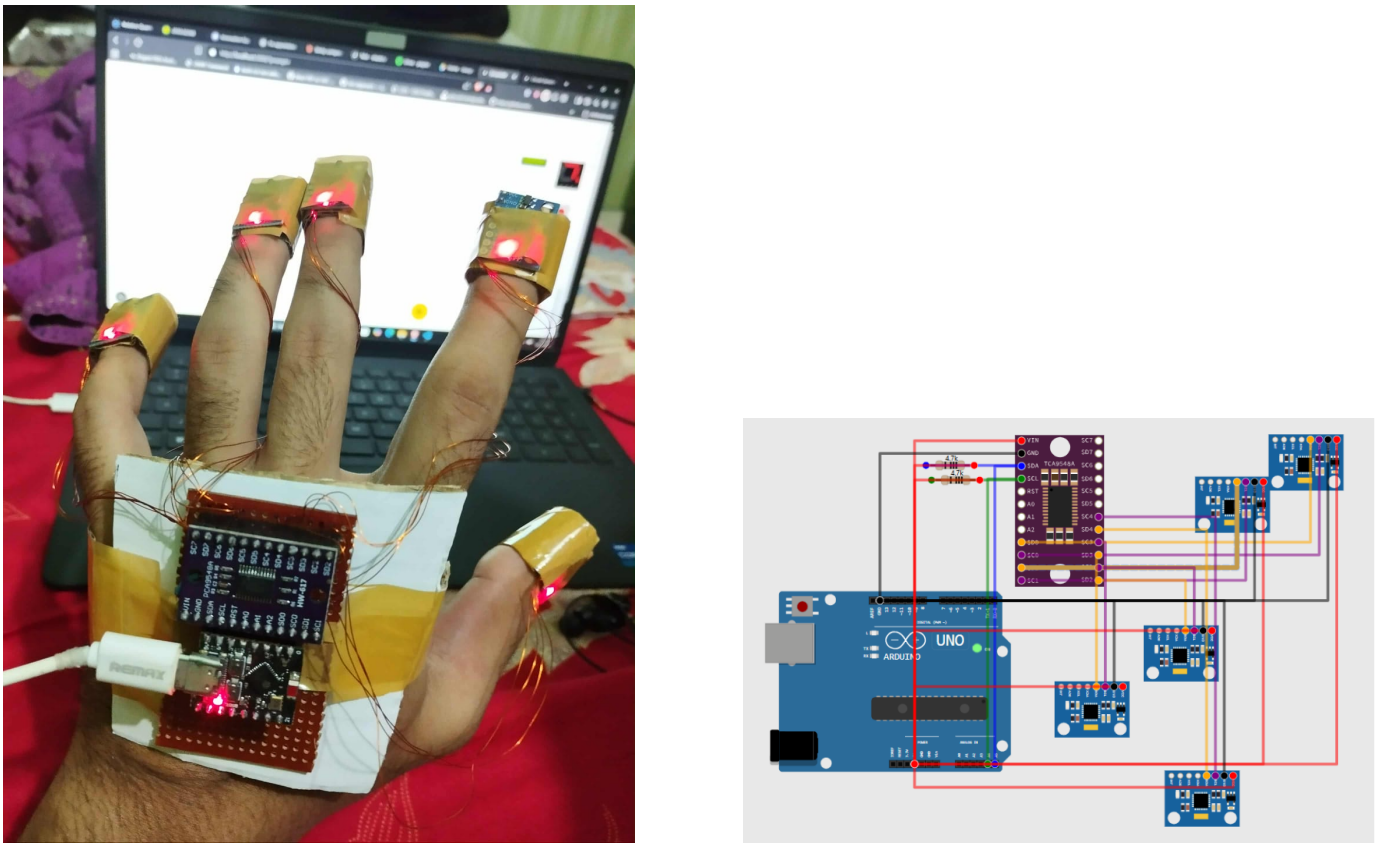


Fig. 1. Left: the physical glove, with one MPU6050 per finger and a sixth reference unit on the back of the hand. Right: circuit schematic showing the six MPU6050 modules multiplexed through the TCA9548A onto a single ESP32 I^2C bus.

B. Wrist-relative orientation

Each finger's orientation relative to the wrist is computed as a full rotation-matrix composition,

$$R_{\text{rel}} = R_{\text{wrist}}^{-1} R_{\text{finger}} = R_{\text{wrist}}^{\top} R_{\text{finger}}, \tag{1}$$

with Euler angles subsequently extracted from R_{rel} for logging. This is a deliberate design choice: naive component-wise subtraction of Euler angles (finger_roll − wrist_roll, etc.) is *not* equivalent to true rotation composition except in degenerate single-axis cases, and produces errors of tens of degrees at realistic hand orientations. We verify this empirically in Section IV-B.

C. Instrument validation

Before any data collection, the following properties were verified against synthetic and physical fault-injection tests:

- **Dead-sensor detection.** A sensor that never wakes returns raw zeros; once the accelerometer's calibrated offset is subtracted, this reads as a deceptively perfect $9.810\,\text{m/s}^2$ gravity magnitude. Detection therefore keys on channel constancy, not on gravity magnitude alone.
- **Multiplexer fault detection.** If the TCA9548A fails to switch, two channels return byte-identical streams; dead sensors are excluded from this comparison first so that two independently-dead sensors are not misreported as a wiring fault.
- **Rotation-math invariance.** Freezing a hand shape while moving the whole arm rigidly must leave relative angles unchanged; this is tested directly (Section IV-B).
- **Channel-to-finger mapping.** Moving one finger at a time and observing which logged column reacts. This procedure identified and corrected a physically crossed Thumb/Index wiring pair.

III. Dataset

The dataset consists of 200 labeled isolated-sign trials (10 classes × 20 repetitions) and 24 unlabeled sessions (12 resting, 12 free hand movement), all recorded from a single signer in one continuous session. Sign classes are Bangla kinship terms (Table II); an eleventh sign ("W20") was dropped during collection because it requires body articulation beyond the hand and forearm, which the glove cannot observe — a scope boundary discovered empirically rather than assumed in advance.

Trial duration ranges 0.9–5.8 s (mean 2.06 s), giving 55–371 samples per trial at the measured sampling rate.

A. Recording-order confound

Signs were recorded in ten contiguous blocks (one block per class) rather than interleaved, over a 284-minute session.

TABLE II
SIGN CLASSES. W6'S ENGLISH GLOSS WAS NOT RECORDED IN THE SESSION LOG; ITS 20 REPETITIONS REMAIN A VALID, INTERNALLY-CONSISTENT, BUT ANONYMOUSLY-LABELED CLASS.

Code	Gloss
W1	Father
W3	Brother
W4	Sister
W5	Wife
W6	(gloss not recorded)
W7	Paternal Aunt
W8	Grandfather
W9	Grandmother
W11	Brother-in-Law
W12	Sister's Husband

TABLE III
SHORTCUT-FEATURE ANALYSIS. CHANCE IS 0.100.

Feature family	Dim.	Accuracy
Yaw drift (a clock)	6	0.430
Trial duration	1	0.295
Yaw drift + duration	7	0.495
Roll/pitch means (hand shape)	12	0.895

This raises the possibility that recording time, not hand shape, drives classification accuracy. This was quantified directly rather than assumed away: features containing no hand-shape information whatsoever — gyroscope yaw drift, which behaves as a monotonic clock, and raw trial duration — were used alone to predict the sign label.

The combined time-based shortcut ceiling is 0.495; legitimate hand-shape features alone reach 0.895, well above that ceiling. All accuracy figures reported below should be read against this bound.

IV. ROTATION-ROBUSTNESS EXPERIMENTS

A. Feature sets

Four feature sets are compared across all experiments:

- **REL_ORI** (75-dim): wrist-relative finger orientation only.
- **ABS_ORI** (90-dim): raw Earth-frame orientation only.
- **ABS+REL** (165-dim): both representations concatenated.
- **FULL_ABS** (270-dim): absolute orientation plus raw accelerometer and gyroscope channels.

Three classifiers are evaluated on each: Random Forest, SVM, and a compact CNN–BiLSTM with temporal attention pooling, under 5-fold cross-validation with macro-F1 as the primary metric.

B. Verifying the invariance claim on logged data

Before any classifier is trained, the invariance premise itself is checked against the actual logged values. For one representative thumb sample, the firmware's stored relative-orientation columns are compared against (a) the explicit matrix computation $R_{\text{wrist}}^{-1} R_{\text{thumb}}$ and (b) naive Euler-angle subtraction (Table IV).

TABLE IV
STORED VS. RECOMPUTED RELATIVE ORIENTATION (ROLL, PITCH, YAW, DEGREES).

Quantity	(roll, pitch, yaw)
Stored `rel_*` columns	$(-6.49,\ 9.57,\ -155.32)$
$R_{\text{wrist}}^{-1} R_{\text{thumb}}$	$(-6.491,\ 9.573,\ -155.319)$
Naive Euler subtraction	$(-26.26,\ -23.38,\ -156.93)$

The stored columns match true matrix composition to three decimal places and disagree with naive subtraction by up to $33°$. A synthetic global rotation was then applied to every logged orientation sample (mathematically equivalent to the signer facing a different direction), and the resulting change in each representation was measured directly: the relative representation changed by at most 5.7×10^{-14} degrees (floating-point noise), while the absolute representation changed by up to $211.09°$. The invariance is therefore confirmed as an exact property of the logged data, not an assumption.

C. RQ1: Clean-condition accuracy

On unperturbed data, representations containing absolute orientation saturate near-perfectly (Table V), because within a single recording session the signer's world-frame heading is nearly constant per sign — an artifact of the recording, not of the gesture.

REL_ORI trails by 6–8 points precisely because it discards this session-specific shortcut.

D. RQ5: Rotation robustness across model families

Applying the same synthetic global rotation used in Section IV-B at test time, macro-F1 is re-measured for every feature set and classifier (Table VI).

REL_ORI loses exactly zero macro-F1 for all three classifiers — consistent with, and a direct empirical confirmation of, the algebraic guarantee. ABS_ORI collapses to near-chance regardless of model capacity, showing that the effect is a property of the representation, not something a sufficiently expressive model can learn around. The one apparent exception, FULL_ABS/Random Forest (drop of only 0.015), is not genuine robustness: it survives by relying on the untouched raw accelerometer/gyroscope channels that the synthetic rotation test does not perturb, a limitation of the stress test rather than evidence of representational robustness.

E. RQ3: Does combining representations help?

A natural question is whether simply concatenating both representations lets a classifier learn to rely on the robust subset. It does not (Table VI, ABS+REL row): all three classifiers still lose 0.77–0.81 macro-F1 under rotation, nearly as much as ABS_ORI alone, despite the exactly-invariant REL_ORI features being present in the same input vector. Every model family, when given the choice, exploits the more discriminative but fragile absolute features on clean training data and inherits their fragility under rotation.

TABLE V
CLEAN-CONDITION MACRO-F1 (5-FOLD CV).

Feature set	Random Forest	SVM	CNN–BiLSTM–Attn
REL_ORI	0.903	0.933	0.907
ABS_ORI	1.000	0.990	0.984
ABS+REL	1.000	0.984	0.990
FULL_ABS	1.000	1.000	1.000

TABLE VI
MACRO-F1 LOST UNDER SYNTHETIC GLOBAL ROTATION (LOWER IS BETTER).

Feature set	Random Forest	SVM	CNN–BiLSTM–Attn
REL_ORI	0.000	0.000	0.000
FULL_ABS	0.015	0.575	0.527
ABS+REL	0.774	0.781	0.806
ABS_ORI	0.815	0.864	0.869

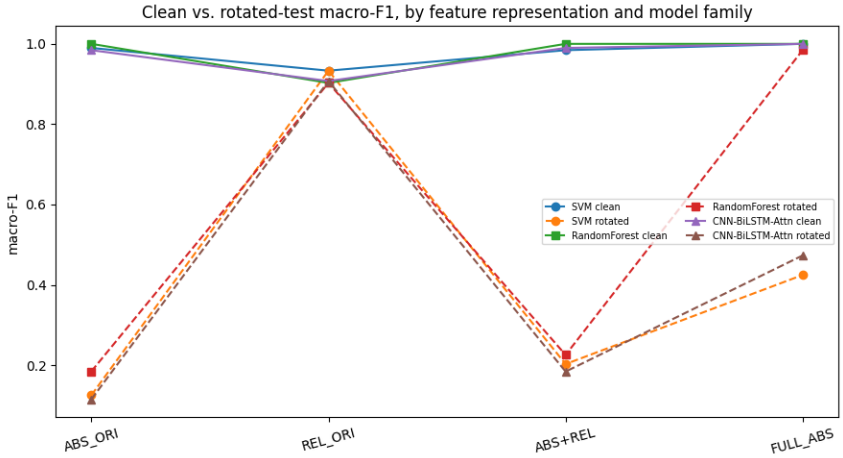


Fig. 2. Clean (solid) vs. rotated-test (dashed) macro-F1 for all three classifiers, across all four feature sets. REL_ORI is the only feature set at which the solid and dashed lines coincide; every representation containing absolute orientation collapses under rotation regardless of classifier.

TABLE VII
ABS_ORI MACRO-F1 LOST UNDER ROTATION, WITH AND WITHOUT ROTATION AUGMENTATION.

Classifier	W/o aug.	With aug.
Random Forest	0.815	0.182
SVM	0.864	0.230
CNN–BiLSTM–Attn	0.869	0.402

F. RQ2: Does rotation augmentation rescue absolute orientation?

Training on synthetically rotated copies of ABS_ORI data substantially reduces — but does not eliminate — the rotation sensitivity (Table VII).

Augmented ABS_ORI still lands at 0.55–0.74 rotated macro-F1, below REL_ORI's constant 0.90–0.93, and only covers the specific rotation distribution sampled during training. REL_ORI's invariance is exact and requires no augmentation.

G. RQ4: Qualitative failure analysis

Because absolute-orientation models saturate at or near 100% clean accuracy, their confusion matrices are uninformative. The REL_ORI/SVM confusion matrix (which sits at 0.933 macro-F1) is analyzed instead (Figure 3). Class W6 — the sign whose gloss was never recorded — is the weakest class, with recall dropping to 0.70 under the neural model and confusions concentrated with W1, W12, and W5. Classes W4, W7, and W8 are recognized perfectly under REL_ORI across all three classifiers.

As a lightweight interpretability check, the CNN–BiLSTM's temporal attention weights were inspected against finger-flexion signal on a handful of trials (Figure 4). Attention does not consistently lock onto the gesture window across trials, and is reported here as a weak diagnostic signal rather than as evidence that the network performs gesture-level reasoning.

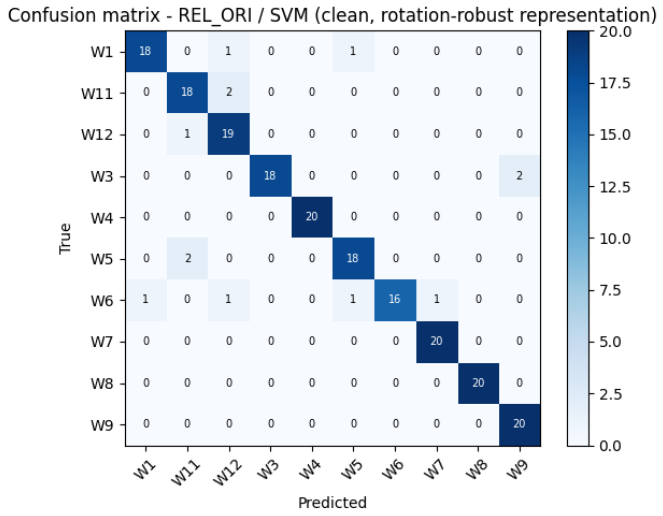


Fig. 3. Confusion matrix, REL_ORI / SVM (macro-F1 = 0.933). Unlike the near-diagonal, uninformative matrices produced by absolute-orientation models, this one shows interpretable structure: W6 is the weakest class, and W11/W12 exchange errors with each other.

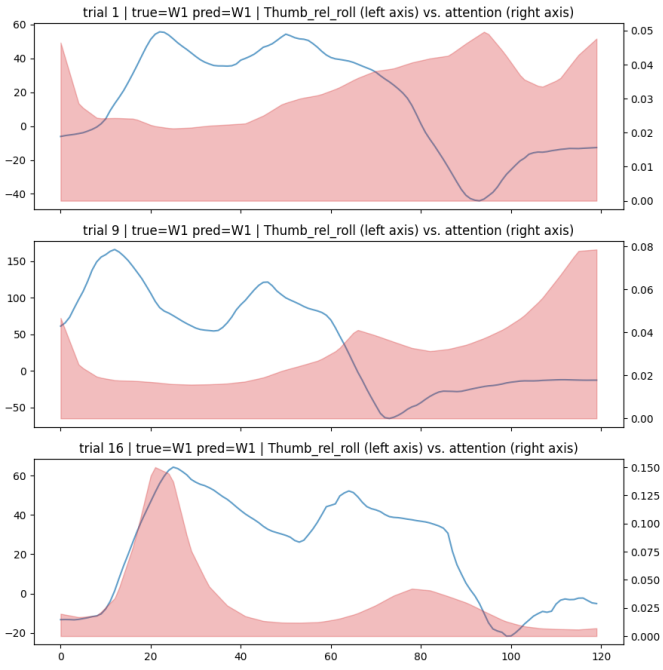


Fig. 4. Temporal attention weights (right axis) against thumb relative-roll signal (left axis) for three W1 test trials, REL_ORI / CNN–BiLSTM–Attn. The network attends closely to the flexion onset in one trial and spreads toward the trial edges in the other two — a weak, not fully consistent, interpretability signal.

V. Self-Supervised Pretraining: A Negative Result

A separate line of work explored whether masked-reconstruction self-supervised pretraining on the 24 unlabeled sessions could reduce the number of labeled examples needed per sign. A 1D-CNN autoencoder was pretrained to reconstruct masked spans of the wrist-relative feature sequence.

TABLE VIII
Masked-reconstruction loss: trained encoder vs. no-learning baselines.

Method	Masked MSE
Guess the channel mean	1.000
Linear interpolation across the gap	1.038
Best linear cross-finger prediction	0.682
Blended no-learning baseline	0.860
Trained SSL encoder (best epoch)	**0.953**

A. The pretext task was too easy

Before trusting any downstream number, the pretext task itself was validated against no-learning baselines evaluated on the identical masked positions (Table VIII). Lower masked-MSE is better.

The trained encoder is outperformed by a no-learning blended baseline by 0.093. An earlier version of this experiment had additionally masked a span narrower than the encoder's receptive field, which reduced the task to interpolation and produced misleadingly encouraging numbers; widening the mask exposed the true difficulty. Given that the encoder cannot beat a method that learns nothing, no downstream label-efficiency claim is made from this pretraining pass. This is reported explicitly as a negative result: at 24 unlabeled sessions from a single signer, neither the corpus scale nor the pretext task difficulty justified the self-supervised machinery, and the diagnosis (rather than the raw negative number) is the useful output of this line of work.

VI. Limitations

1) **Single signer, single continuous session.** All results support within-session generalization claims only. No cross-signer or cross-day evidence exists.
2) **Synthetic rotation, not physical.** The rotation-robustness test applies an exact mathematical $SO(3)$ rotation to logged orientation values. It does not model glove slippage, sensor-to-finger remounting, or magnetometer/heading estimation error under physically rotated recording conditions.
3) **Not a like-for-like benchmark comparison.** Published data-glove results (e.g. WaveGlove, the LSF glove) use different sensors, vocabularies, and evaluation splits; they are cited as context, not as numbers to match or exceed.
4) **The neural network is a compact baseline**, sized for approximately 160 training trials per fold, and is not tuned as aggressively as the classical models; it is included to test whether the representation effect generalizes across model families, not to claim state-of-the-art performance.
5) **Recording-order confound**, quantified in Section III-A but not eliminated by design; interleaving repetitions in future recording sessions would remove it at the source.

6) **Measured sampling rate (62.5–67.8 Hz)** falls short of the 100 Hz firmware target, attributed to I^2C read latency across six multiplexed sensors.

VII. Conclusion

Reporting finger orientation relative to a wrist-mounted reference IMU, computed via true rotation-matrix composition rather than Euler-angle subtraction, produces a representation with a provable and empirically confirmed invariance to whole-arm rotation. This invariance reproduces identically across three structurally different classifiers (Random Forest, SVM, and a CNN–BiLSTM–attention network), each losing exactly zero macro-F1 under a synthetic global rotation stress test, against a loss of 0.82–0.87 for absolute-orientation representations under the identical test. The cost of this guarantee is 6–8 points of clean, within-session macro-F1 on the recorded dataset. Rotation augmentation of absolute-orientation features narrows but does not close this robustness gap, and simply concatenating both representations does not allow a classifier to recover the invariance on its own. A companion self-supervised pretraining experiment is reported as an explicit negative result at the current data scale. The primary limitation throughout is that all findings are within-signer; recruiting additional signers is the single most important next step for testing whether these conclusions generalize.